FINITE TYPE AFTER REDUCTION IS STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof identifies the reduction of the completed tensor product as the ordinary base change of the reductions and applies stability of finite type under base change

Editorial note. The label lemma-base-change-finite-type-red names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the property P on arrows of WAdm^count defined in lemma-finite-type-red of the same chapter: B/b -> A/a is of finite type for the ideals of topologically nilpotent elements
- weak admissibility makes the ideals of topologically nilpotent elements open
- the completed tensor product of weakly admissible rings and its kernel being a weak ideal of definition (compare Formal Spaces, Lemma formal-spaces-lemma-completed-tensor-product)
- finite type of ring maps is stable under base change

Source reference. Original lemma 0GB4: sources/stacks-chow/tags.txt, source lines 20359–20359. Prerequisite pointer invoked at proof line(s) 2719; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GBC: sources/stacks-chow/tags.txt, source lines 20367–20367. Prerequisite pointer invoked at proof line(s) 2701; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GC2: sources/stacks-restricted/pinned-original-source.tex, source lines 2631–2631. Prerequisite pointer invoked at proof line(s) 2699; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Consider the property P on arrows of $WAdm^{count}$ defined in Lemma 0GC2. Then P is stable under base change (Formal Spaces, Situation 0GBC).*

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The statement makes sense by Lemma 0GC2. To see that it is true assume we have morphisms $B \to A$ and $B \to C$ in $WAdm^{count}$ such that $B/\mathfrak{b} \to A/\mathfrak{a}$ is of finite type where $\mathfrak{b} \subset B$ and $\mathfrak{a} \subset A$ are the ideals of topologically nilpotent elements. Since A and B are weakly admissible, the ideals $\mathfrak{a}$ and $\mathfrak{b}$ are open. Let $\mathfrak{c} \subset C$ be the (open) ideal of topologically nilpotent elements. Then we find a surjection $A\widehat{\otimes}_B C \to A/\mathfrak{a} \otimes_{B/\mathfrak{b}} C/\mathfrak{c}$ whose kernel is a weak ideal of definition and hence consists of topologically nilpotent elements (please compare with the proof of Formal Spaces, Lemma 0GB4). Since already $C/\mathfrak{c} \to A/\mathfrak{a} \otimes_{B/\mathfrak{b}} C/\mathfrak{c}$ is of finite type as a base change of $B/\mathfrak{b} \to A/\mathfrak{a}$ we conclude. $\square$